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United States Patent Application Publication

20250285804

Kind Code

A1

Publication Date

September 11, 2025

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COIL DEVICE

Abstract

A coil device including: a first conductor having a first main body, a second conductor having a second main body in the first main body and extending along the first main body, and a core in which the first and the second main body are inside of the core; wherein the second main body is inside of the first main body in the core and extends along the first main body, the second conductor includes a second mounting part formed with the second main body, a first mounting part of the first conductor and the second mounting part of the second conductor along a base surface of the core, and a second lead-out part between the second main body and the second mounting part includes a curved part which changes a lead-out direction so that the second mounting part is pulled out to a position different from the first.

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Appl. No.: 19/057583

Filed: February 19, 2025

Foreign Application Priority Data

JP	2024-037414	Mar. 11, 2024
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Publication Classification

Int. Cl.: H01F27/30 (20060101); H01F27/24 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a coil device, for example, used as an inductor or so.

BACKGROUND

[0002] For example, a coil device shown in the below described Patent Document 1 is proposed. A conventional coil device has a risk of causing short circuit since the distance between mounting parts of a coil is short; and as a measure to correspond to this matter, a structure provided with a resin spacer between the coils is proposed. [0003] [Patent Document 1] JP Patent Laid Open No. 2022-33703

SUMMARY

[0004] The object of the present disclosure is to provide a coil device capable of reducing short circuit malfunctions between conductors configuring the coil without using a resin spacer.

[0005] In order to achieve the above-mentioned object, the coil device according to one embodiment of the present disclosure includes: [0006] a first conductor having a first main body, [0007] a second conductor having a second main body arranged inside the first main body and at least partially extending along the first main body, and [0008] a core in which the first main body and the second main body are arranged inside of the core; [0009] wherein [0010] the second main body is arranged inside of the first main body in the core and extends along the first main body, [0011] the second conductor includes a second mounting part continuous with the second main body, [0012] a first mounting part of the first conductor and the second mounting part of the second conductor are arranged along a base surface of the core, and [0013] a second lead-out part provided between the second main body and the second mounting part includes a curved part which changes a lead-out direction along the base surface so that the second mounting part is pulled out to a position different from the first mounting part.

[0014] In this coil device, the second main body of the second conductor is arranged inside of the first main body of the first conductor in the core, and the second main body and the first main body are stacked (overlying) while taking a predetermined space in between. Thus, efficient transfer of magnetic flux between the first conductor and the second conductor is possible, and magnetic coupling between the first conductor and the second conductor can be enlarged sufficiently.

[0015] Also, in this coil device, the second lead-out part has a curved part which changes a lead-out direction along the base surface of the core. Thus, it is possible to prevent short circuits between the end parts of the second conductor and also possible to effectively prevent short circuits between the first conductor and the second conductor without using a resin spacer.

[0016] The second lead-out part may be configured of the curved part only; however, the second lead-out part may further have an intermediate part which connects the curved part and the second mounting part in a continuous manner. The intermediate part preferably has a straight-line edge which is continuous with an outline of the curved part; however, the edge may be a curved line, a zigzag-line, or a wavy form line.

[0017] Preferably, the second lead-out part includes a pair of second lead-out parts formed at both ends of the second conductor, [0018] each of the pair of the second lead-out parts respectively comprises the straight-line edge, [0019] the straight-line edge of one of the pair of second lead-out parts and the straight-line edge of the other one of the pair of second lead-out parts have a predetermined distance between each other and are arranged parallel to each other, and [0020] the second mounting part includes a pair of second mounting parts and each of the pair of second mounting parts is arranged at a position which is opposite with respect to each other along the base

surface of the core.

[0021] By configuring as such, an insulation distance between the pair of straight-line edges can be secured easily; hence, even if the area of the base surface is relatively small, the second lead-out parts positioned at the both ends of the second conductor can be easily insulated securely. Also, the mounting parts positioned at the both ends of the second conductor can be easily insulated securely. Furthermore, insulation between the first mounting part and the second mounting part can be secured easily.

[0022] The core may include a first core and a second core facing the first core. Preferably, the second mounting part is positioned outside of the first mounting part along a first direction parallel to a facing direction of the first core and the second core on the base surface of the core.

[0023] By configuring as such, even if the area of the base surface of the core is relatively small, the second lead-out parts positioned at the both ends of the second conductor can be easily insulated securely. Also, the second mounting parts positioned at the both ends of the second conductor can be easily insulated securely. Further, the first mounting part and the second mounting part can be easily insulated securely.

[0024] Preferably, a conductor width of the second mounting part is wider than a conductor width of the second lead-out part. By configuring as such, a bonding area of the second mounting part and a land pattern of a circuit board or so increases and a bonding strength improves. Also, even if positions between the land pattern of the circuit board or so and the second mounting part are shifted, the bonding strength between the land pattern and the second mounting part can be secured easily.

[0025] Preferably, the first mounting part includes a pair of first mounting parts formed at both ends of the first conductor. Also, preferably, the second mounting part includes a pair of second mounting parts formed at both ends of the second conductor. A base surface of the core may include at least four outer edges configuring a polygon shape. Preferably, each of the pair of first mounting parts is arranged in a direction facing two of the at least four outer edges facing away from each other, and

[0026] each of the pair of second mounting parts is arranged in a direction facing two of the at least four outer edges facing away from each other which are different edges that the first mounting part are facing.

[0027] By configuring as such, even if the area of the base surface is relatively small, the second lead-out parts which are positioned at the both ends of the second conductor can be easily insulated securely. Also, the second mounting parts which are positioned at the both ends of the second conductor can be easily insulated securely. Further, the first mounting part and the second mounting part can be easily insulated securely.

[0028] Preferably, the first conductor and the second conductor each respectively have a flat wire shape within one turn.

[0029] Preferably, a width of the second main body is narrower than a width of the first main body. When such relation is satisfied, the curved part can be easily formed to the second lead-out part; and, even if the area of the base surface is relatively small, the second lead-out parts positioned at both ends of the second conductor can be easily insulated securely. Also, the second mounting parts which are positioned at the both ends of the second conductor can be easily insulated securely. Further, the first mounting part and the second mounting part can be easily insulated securely.

[0030] An inner surface of the core may include a step part corresponding to each width of the second main body and the first main body.

[0031] Preferably, an insulation layer may be formed on a surface of the first main body. An insulation layer may not be formed on the surface of the second main body.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0032] FIG. 1 is a perspective view of a coil device according to one embodiment of the present disclosure.

[0033] FIG. 2A is an explosive perspective view of the coil device shown in FIG. 1.

[0034] FIG. 2B is an explosive perspective view of the coil device shown in FIG. 1, and an observation angle is different from FIG. 2A.

[0035] FIG. 3A is a bottom view of the coil device shown in FIG. 1.

[0036] FIG. 3B is a bottom view of a coil device according to further another embodiment.

[0037] FIG. 3C is a bottom view of a coil device according to further another embodiment.

[0038] FIG. 3D is a bottom view of a coil device according to further another embodiment.

[0039] FIG. 4 is a bottom view of the coil device shown in FIG. 1.

[0040] FIG. 5 is a perspective view of the base surface showing an arrangement method according to one example of the coil device shown in FIG. 1.

DESCRIPTION

[0041] In below, embodiments of the present disclosure are described.

First Embodiment

[0042] As shown in FIG. 1, a coil device **10** according to the present embodiment is approximately a rectangular parallelepiped shape; and, it functions as a coupling coil used for a power circuit or so. An X-axis direction width of the coil device **10** is preferably 3.0 to 12.0 mm, and a Y-axis direction width is preferably 3.0 to 6.0 mm, and a Z-axis direction width is preferably 3.0 to 12.0 mm. Note that, the X-axis, the Y-axis, and the Z-axis are approximately perpendicular to each other. In the present embodiment, the Z-axis is a height direction of the coil device **10**. Also, in the below description, an inner side is a direction towards the center of a member (for example, the center of the coil device **10**) along each axis, and an outer side is a direction away from the center.

[0043] As shown in FIG. 2A and FIG. 2B, the coil device **10** includes a core **20** configured of a first core **20a** and a second core **20b**, a first conductor **30**, and a second conductor **40**. Either one of the first conductor **30** and the second conductor **40** functions as a primary coil, and the other one functions as a secondary coil. Details of the conductors **30** and **40** are described later.

[0044] The first core **20a** and the second core **20b** respectively has a configuration of a so-called E-type core; and preferably these have the same shapes; however, the shapes do not necessarily have to be the exact same. For example, a length of an outer leg part **22a** and **22b** of the core **20a** and **20b**, which are described later, along the X-axis may be different and a length of a middle leg part **23a** and **23b** along the X-axis may be different. Also, the length of the either one of the outer leg part **22a** or the outer leg part **22b** may be made longer, and a length of the other one of the outer leg part **22a** or the outer leg part **22b** may be zero.

[0045] In the present embodiment, each core **20a** and **20b** respectively has the same configuration as described in below.

[0046] The first core **20a** includes a first base part **21a** having an outer side surface **20a3** extending along the Y-axis and the Z-axis, and a pair of outer leg parts **22a** and **22a** which protrude inward along the X-axis from the both ends along the Y-axis of the first base part **21a**. When the core **20a** and the core **20b** are combined, the outer side surfaces **20a4** and **20a4** of each outer leg part **22a** and **22a** are positioned roughly on the same plane as the outer side surface **20b4** of the second core **20b** which are described later. Here, "roughly on the same plane" means that a production error, or ± 0.2 mm or so of shifting is permissible. The same applies hereinafter.

[0047] Also, at the inside of the first base part **21a** which is positioned at the center in the Y-axis direction of the pair of outer leg parts **22a** and **22a**, a middle leg part **23a** is integrally formed which protrudes inwards from the base part **21a**. At a tip side along the X-axis direction of the middle leg part **23a**, a step-like depressed part **24a** is respectively formed at the both ends along the Y-axis; and, a first raised part **25a** is formed between the step-like depressed parts **24a**. The first

raised part **25a** is a part which enters inside the second main body **41** of the second conductor **40**.
[0048] The first step-like depressed part **24a** engages with one of the side edges of a leg part **41a** extending along the Z-axis of the second main body **41**. Also, a connecting part **41b** integrally connects an upper end in the Z-axis direction of a pair of leg parts **41a** of the second main body **41**; and the connecting part **41b** is disposed on an upper end face **25a1** of the raised part **25a** of the core **20a**.

[0049] A first groove **26a** is formed between the first middle leg part **23a** and the first outer leg part **22a** of the first core **20a**. One of side edges of a leg part **31a** extending along the Z-axis of the first main body **31** of the first conductor **30** is inserted into and engages with the first groove **26a**. Also, a connecting part **31b** integrally connects upper ends in the Z-axis of a pair of leg parts **31a** of the first main body **31**; and the connecting part **31b** is disposed on an upper end surface **23a1** of the middle leg part **23a** of the core **20a**.

[0050] The upper end surface **23a1** of the middle leg part **23a** is one step lower along the Z-axis than the outer leg part **22a** of the core **20a** and the upper end surface **20a1** of the base part **21a**. The upper end surface **25a1** of the raised part **25a** is one step lower along the Z-axis than the upper end surface **23a1** of the middle leg part **23a**. A height difference along the Z-axis between the upper end surface **23a1** of the middle leg part **23a** and the upper end surface **20a1** of the core **20a** is preferably about the same as the thickness of the connecting part **31b** of the first conductor **30**, and also preferably the upper surface of the connecting part **31b** and the upper end surface **20a1** are positioned on about the same plane. Also, a height difference along the Z-axis between the upper end surface **25a1** of the raised part **25a** and the upper end surface **23a1** of the middle leg part **23a** is preferably about the same or thicker than the thickness of the connecting part **41b** of the second conductor **40**.

[0051] The second core **20b** has a second base part **21b** including an outer side surface **20b3** which extends along the Y-axis and the Z-axis, and the pair of outer leg parts **22b** and **22b** which protrudes inward along the X-axis from both ends along the Y-axis of the second base part **21b**. When the core **20a** and the core **20b** are combined, the outer side surfaces **20b4** and **20b4** of the outer leg part **22b** are positioned on about the same plane as the outer side surfaces **20a4** and **20a4** of the aforementioned first core **20a**.

[0052] Also, at the inner surface of the second base part **21b** positioned at the center part in the Y-axis of the pair of outer leg parts **22b** and **22b**, a middle leg part **23b** is integrally formed which protrudes inward from the base part **21b**. At a tip side along the X-axis direction of the middle leg part **23b**, a step-like depressed part **24b** is respectively formed at the both ends along the Y-axis; and, a second raised part **25b** is formed between the step-like depressed parts **24b** and **24b**. The second raised part **25b** is a part which protrudes along the X-axis and enters inside the second main body **41** of the second conductor **40**.

[0053] The second step-like depressed part **24b** engages with the other one of the side edges of leg part **41a** extending along the Z-axis of the second main body **41**. Also, the connecting part **41b** integrally connects an upper end of a pair of leg parts **41a** in the Z-axis direction of the second main body **41**; and the connecting part **41b** is disposed over an upper end face **25b1** of the raised part **25b** of the core **20b**.

[0054] A second groove **26b** is formed between the second middle leg part **23b** and the second outer leg part **22b** of the second core **20b**. The other one of the side edges of the leg part **31a** extending along the Z-axis of the first main body **31** of the first conductor **30** is inserted into and engages with the second groove **26b**. Also, the connecting part **31b** integrally connects upper ends of a pair of the leg parts **31a** in the Z-axis of the first main body **31**; and the connecting part **31b** is disposed over an upper end surface **23b1** of the middle leg part **23b** of the core **20b**.

[0055] As shown in FIG. 2B, the upper end surface **23b1** of the middle leg part **23b** is one step lower along the Z-axis than the outer leg part **22b** of the core **20b** and the upper end surface **20b1** of the base part **21b**. The upper end surface **25b1** of the raised part **25b** is one step lower along the Z-

axis than the upper end surface **23b1** of the middle leg part **23b**. A height difference along the Z-axis between the upper end surface **23b1** of the middle leg part **23b** and the upper end surface **20b1** of the core **20b** is preferably about the same as the thickness of the connecting part **31b** of the first conductor **30**, and also preferably the upper surface of the connecting part **31b** and the upper end surface **20b1** are positioned on about the same plane. Also, a height difference along the Z-axis between the upper end surface **25b1** of the raised part **25b** and the upper end surface **23b1** of the middle leg part **23b** is preferably about the same or thicker than the thickness of the connecting part **41b** of the second conductor **40**.

[0056] The tip surface along the X-axis of the outer leg part **22a** of the first core **20a** and the tip surface along the X-axis of the outer leg part **22b** of the second core **20b** preferably face against each other and bonded using an adhesive or so. The tip surface along the X-axis of the middle leg part **23a** of the first core **20a** (specifically, the tip surface of the raised part **25a**) and the tip surface along the X-axis of the middle leg part **23b** of the second core **20b** (specifically, the tip surface of the raised part **25b**) may face against each other and adhered, or there may be a predetermined gap in between.

[0057] The first core **20a** and the second core **20b** are each configured of a magnetic material, and for example, it may be a magnetic material with a relatively high permeability. Examples of such magnetic material include ferrite such as Ni—Zn-based ferrite, and Mn—Zn-based ferrite; and a metal magnetic material. Also, the first core **20a** and the second core **20b** may be each configured of a sintered body, or it may be formed by molding a resin including a magnetic powder.

[0058] The first conductor **30** is configured of a conductor plate of a flat plate shape (such as a flat wire), and the first conductor **30** includes the first main body **31** which is approximately a U-shape. The first conductor **30** is arranged, together with the second conductor **40**, between the first core **20a** and the second core **20b**. Examples of the material configuring the first conductor **30** include good conductors such as copper and copper alloy, silver, nickel, etc.; and as long as the material is a conductive material, it is not particularly limited. For example, the first conductor **30** is formed by mechanically processing a metal plate, and a method for forming the first conductor **30** is not limited to this.

[0059] In the example shown in the figure, the first conductor **30** has an elongated shape as a whole, and a height along the Z-axis of the first conductor **30** is larger than widths in the X-axis direction and the Y-axis direction. The cross-section area perpendicular to an extending direction of the first conductor **30** is preferably about the same as the cross-section area perpendicular to an extending direction of the second conductor **40**. The thickness (the plate thickness) of the first conductor **30** is thinner by about $1/n$ than the thickness (the plate thickness) of the second conductor **40**, and a width along the X-axis of the first conductor **30** is preferably wider by about n times than a width along the X-axis direction of the second conductor **40**. Note that, n is preferably any number of 2 or larger. The thickness of the first conductor **30** is preferably 0.3 to 2.0 mm.

[0060] A plating layer may be formed on the entire surface of the first conductor **30**. The plating layer may be a single layer or a plurality of layers; and for example, the plating layer may be configured of Cu plating, Ni plating, Sn plating, Ni—Sn plating, Cu—Ni—Sn plating, Ni—Au plating, Au plating, etc. The plating layer is formed on the surface of the first conductor **30** by carrying out an electroplating or electroless plating. A thickness of the plating layer is not particularly limited, and preferably it is 1 to 30 μm .

[0061] The first conductor **30** has a first main body **31** and first mounting parts **32** and **32** which are respectively formed integrally at both ends of the first main body **31**. The first main body **31** has a pair of leg parts **31a** extending in the Z-axis direction while taking a predetermined distance along the Y-axis direction, and the first main body **31** also has the connecting part **31b** which integrally connects the upper ends of the leg parts **31a**. The first main body **31** is a coil formed approximately in a U-like shape of less than 1 turn.

[0062] A lead-out part **33** is formed with a plate surface curved part which the angle of the plate

surface of the conductor **30** bends approximately perpendicularly, and the plate surface curved part is provided between the mounting part **32** and lower ends of each of the leg parts **31a** of the first main body **31**. Either one of the pair of mounting parts **32** functions as an input terminal (or an output terminal) and the other one functions as an output terminal (or an input terminal). The mounting parts **32** and **32** extend outward along the Y-axis. For example, the mounting parts **32** and **32** connect to a first land electrode **53** formed on the mounting surface of the base board **50** for mounting as shown in FIG. 5 via a bonding member such as a solder, a conductive adhesive agent, etc.

[0063] As shown in FIG. 1, the mounting parts **32** and **32** are exposed to the outside from the base surfaces **20a2** and **20b2** of the core **20**. Also, the insulation coating which coats the first conductor **30** is removed from at least the lower surface of the mounting parts **32** and **32**, and the first land electrode **53** shown in FIG. 5 or so and the first conductor **30** can be connected electrically.

[0064] As shown in FIG. 2A and FIG. 2B, the second conductor **40** is formed, for example, by press processing a metal plate; and, the second conductor **40** includes the second main body **41** which is approximately a U-shape, the lead-out parts **43** formed at the both ends of the second main body **41**, and the mounting part **42** formed at the tip of the lead-out part **43**. The second conductor **40** can be configured of the same material as the first conductor **30**; however, the second conductor **40** has the lead-out part **43** of a complex configuration, unlike the lead-out part **33** of the first conductor **30**; thus, the insulation coating may not be formed to the second conductor **40**.

[0065] The second conductor **40** is arranged inside of the cores **20a** and **20b** as similar to the first conductor **30** (the second conductor **40** is arranged in the groove parts **26a** and **26b**, and the step-like depressed parts **24a** and **24b**). The leg part **31a** of the first conductor **30** is inserted in the groove parts **26a** and **26b**, and the leg parts **41a** and **41a** of the second conductor **40** engage with step-like depressed parts **24a** and **24b**. By arranging as such, the second main body **41** of the second conductor **40** is arranged inside the first main body **31** of the first conductor **30** by taking a predetermined space. Note that, in the case that the surface of the first main body of the first conductor **30** is insulation coated, the second main body **41** may contact to at least part of the inner side of the first main body **31**.

[0066] In the example shown in the figures, the second conductor **40** has an elongated shape, and a height in the Z-axis direction of the second conductor **40** is longer than the length in the Y-axis direction of the second conductor **40**. The second main body **41** of the second conductor **40** is smaller than the first main body **31** of the first conductor **30**, and when these are arranged, the second main body **41** is surrounded by the first main body **31**. The width along the X-axis of the second main body **41** is narrower than the width along the X-axis of the first main body **31**; and the second main body **41** is preferably arranged roughly at the center position along the X-axis of the first main body **31**; however, the position may be shifted slightly.

[0067] The second conductor **40** has the pair of mounting parts **42** formed at the both ends of a belt-like metal plate configuring the second main body **41**. The second main body **41** has a pair of leg parts **41a** extending along the Z-axis direction while taking a predetermined distance along the Y-axis between the leg parts **41a**, and the connecting part **41b** integrally connecting the upper ends of the leg parts **41a**. Further, the second main body **41** configures a coil of approximately a U-shape of less than 1 turn.

[0068] The lead-out part **43** is formed between the mounting part **42** and the lower end of the leg part **41a** of the second main body **41**. Either one of the pair of mounting parts **42** functions as an input terminal (or an output terminal), and the other one functions as an output terminal (or input terminal). The lead-out direction of the mounting parts **42** and **42** is changed by the lead-out parts **43**, and the mounting parts respectively extends along the X-axis.

[0069] These mounting parts **42** and **42** connect to a second land electrode **54a** or a common second land electrode **54b**, for example, formed on the mounting surface of the base board **50** for mounting as shown in FIG. 5 via a bonding member such as a solder, a conductive adhesive agent,

etc.

[0070] As shown in FIG. 1, the lead-out parts **43** and the mounting parts **42** and **42** are exposed to the outside from the base surfaces **20a2** and **20b2** of the core **20**. Also, in the case that the second conductor **40** is insulation coated, the insulation coating is removed from the lower surfaces of the mounting parts **42** and **42**, and the second conductor **40** can be electrically connected to the second land electrode **54a** or a common second land electrode **54b** shown in FIG. 5.

[0071] As shown in FIG. 2A and FIG. 2B, the pair of lead-out parts **43** of the second conductor **40** respectively has a first curved part **43a** corresponding to the lead-out part **33** of the first conductor **30**. The first curved part **43a** is configured of a plate surface curved part where an angle of the plate surface is bent approximately perpendicularly. Unlike the first lead-out part **33** which is bent outward, the plate surface of the lead-out part **43** is bent inward. Due to the first curved part **43a**, the lead-out direction of the lead-out part **43** changes to inward direction along the Y-axis from the lower end of second leg part **41a** shown in FIG. 2A.

[0072] As shown in FIG. 3A and FIG. 4, each of the pair of lead-out parts **43** of the second conductor **40** includes the second curved part **43b** and the intermediate part **43c** in addition to the first curved part **43a**. The second curved part **43b** is continuous with the first curved part **43a**. The lead-out direction of the lead-out part **43** which has been changed by the first curved part is further changed by the second curved part **43b** in a predetermined angle such as an obtuse angle (the angle may be a perpendicular angle or an acute angle) along the base surfaces **20a2** and **20b2** of the core **20**. Note that, unlike the first curved part **43a**, the second curved part **43b** does not change the angle of the plate surface but rather changes the lead-out direction in parallel along the plate surface.

[0073] As shown in FIG. 4, the lead-out direction of one of the lead-out parts **43**, which the lead-out direction has been changed inward along the Y-axis, is changed by one of the pair of second curved parts **43b** in a predetermined angle such as an obtuse angle (the angle may be a perpendicular angle or an acute angle) towards an outer side surface **20a3** of the core **20**. Also, the lead-out direction of the other one of the lead-out parts **43**, which the lead-out direction has been changed inward along the Y-axis, is changed by the other one of the pair of second curved parts **43b** in a predetermined angle such as an obtuse angle (the angle may be a perpendicular angle or an acute angle) towards an outer side surface **20b3** of the core **20**.

[0074] In the present embodiment, the outer edge of the second curved part **43b** is an angular edge when observed from a direction parallel to the Z-axis; however, it is not limited to this and the outer edge may be curved. The intermediate part **43c** is positioned between the second curved part **43b** and the second mounting part **42**, and the intermediate part **43c** has a straight-line edge **43c1** which is continuous with the outline of the second curved part **43b**.

[0075] The straight-line edges **43c1** of the pair of intermediate parts **43c** are parallel to each other and spaced apart by a predetermined distance ($2 \times D1$) along the base surfaces **20a2** and **20b2** of the core **20**; and the straight-line edges **43c1** respectively extends diagonally toward the side surfaces **20a3** and **20b3** positioned opposite to each other. The mounting parts **42** and **42** are integrally formed to the tip of each intermediate part **43c**.

[0076] Regarding one of the pair of mounting parts **42** and **42**, the end edge along the X-axis is positioned at approximately the same plane as the outer side surface **20a3** of the core **20**, and the end edge preferably is positioned approximately at the center of the outer side surface **20a3** along the Y-axis. Also, regarding the other one of the pair of mounting parts **42** and **42**, the end edge along the X-axis is positioned at approximately the same plane as the outer side surface **20b3** of the core **20**, and the end edge is preferably positioned approximately at the center of the outer side surface **20b3** along the Y-axis.

[0077] In the present embodiment, a width **W0** which is perpendicular to the diagonal lead-out direction of the intermediate part **43c** is preferably uniform along the diagonal lead-out direction, and for example, the width **W0** is preferably about the same as a width **W1** along the X-axis of the leg part **41a** and the connecting part **41b** shown in FIG. 2A. The cross-section area of the leg part

41a and the cross-section area of the intermediate part **43c** which are perpendicular to the direction of current flow of the second conductor **40** are roughly the same. Further, the cross-section area of the connecting part **41b** and the cross-section area of the intermediate part **43c** which are perpendicular to the direction current flow of the second conductor **40** are roughly the same. Thereby, uniform electric resistance can be achieved.

[0078] The width along the Y-axis of the mount part **42** is preferably wider than the width perpendicular to the diagonal lead-out direction of the intermediate part **43c**. This is because, as shown in FIG. 5, the mounting parts **42** connect with the land electrodes **54a** and **54b** of the base board **50**; thus, such configuration is preferable from the point of achieving a good electrical connection between the mounting parts **42** and the land electrodes **54a** and **54b**, and also achieving enhanced connection strength.

[0079] The angle of the straight-line edge **43c1** of the lead-out part **43** with respect to the Y-axis (that is, the bent angle of the second curved part **43b**) is determined for example as described in below.

[0080] As shown in FIG. 4, one end along the X-axis of one of inward bending start lines **43a1** of the pair of first curved parts **43a** and **43a** and the other end along the X-axis of the other one of the inward bending start lines **43a1** of the pair of first curved parts **43a** and **43a** are connected using a hypothetical line. Also, a center line which divides a total area of the base surfaces **20a2** and **20b2** of the core **20** into half along a hypothetical line parallel to the Y-axis is defined as a center line **L2**.

[0081] A line crossing the intersection between the center line **L2** and the hypothetical line **L1** and being perpendicular to the hypothetical line **L1** is defined as a hypothetical line **L3**. The angle of the hypothetical line **L3** with respect to the Y-axis is a curved angle of the second curved part **43b**. The straight-line edge **43c1** positioned at the inside of each of the pair of intermediate parts **43c** is set to be parallel to the hypothetical line **L3**. A distance **D1**, which is between the hypothetical line **L3** and the straight-line edge **43c1**, is preferably 0.2 mm or larger, further preferably 0.3 mm or larger, or even more preferably 0.5 mm or larger. Also, a distance **L4** between the curved parts **43a** and **43b** and the second mounting part **42** which is in close proximity with the curved parts **43a** and **43b** is not particularly limited, and preferably it is 0.5 mm or larger, more preferably 0.8 mm or larger, or further preferably 1.0 mm or larger.

[0082] In the present embodiment, the outer side surface **20a3** and the outer side surface **20b3** are positioned facing away from each other along the X-axis of the core **20**. Also, the outer side surface **20a4** and the outer side surface **20a4** are positioned facing away from each other along the Y-axis of the core **20**, and the outer side surface **20b4** and the outer side surface **20b4** are positioned facing away from each other along the Y-axis of the core **20**.

[0083] In the present embodiment, one of the first mounting parts **32** is arranged so that the outer end edge of the first mounting part **32** roughly matches the position corresponding to the plane of one of the outer side surfaces **20a4** and **20b4**, and the other one of the first mounting parts **32** is arranged so that the outer end edge of the first mounting part **32** roughly matches the position corresponding to the plane of one of the outer side surfaces **20a4** and **20b4**. Note that, as similar to “roughly on the same plane”, “roughly matches” means that a production error, or ± 0.2 mm or so of shifting is allowed.

[0084] That is, in the present embodiment, the four mounting parts **32**, **32**, **42**, and **42** are arranged along the base surfaces **20a2** and **20b2** of the core **20** so that the outer end edges of the four mounting parts roughly match to the respective side surface of the four side surfaces of the core **20**. The main body of the mounting part may be in contact with or not in contact with the base surface of the core, and preferably it is in contact.

[0085] In the coil device **10** according to the present embodiment, the second main body **41** of the second conductor **40** is arranged inside the first main body **31** of the first conductor **30** at the inside of the core **20**. The first main body **31** is stacked (overlies) on the second main body **41** by taking a predetermined distance (space or by having the insulation coating) between each other. Therefore,

magnetic flux can be transferred efficiently between the first conductor **30** and the second conductor **40**, and magnetic coupling between the first conductor **30** and the second conductor **40** can be made sufficiently stronger.

[0086] Also, in this coil device **10**, each of the pair of second lead-out parts **43** has the second curved part **43b** which changes the lead-out direction along the base surfaces **20a2** and **20b2** of the core **20**. Thus, short circuit between the terminals of the second conductor **40** or short circuit between the first conductor **30** and the second conductor **40** can be secured effectively without using a resin spacer.

[0087] Each of the pair of second lead-out parts **43** may be configured only from the first curved part **43a** and the second curved part **43b**. However, in the present embodiment, the second lead-out part **43** further includes the intermediate part **43c** which connects the second curved part **43b** and the second mounting part **42** in a continuous manner. Each of the pair of intermediate parts **43c** includes the straight-line edge **43c1** which is continuous with the outline of the second curved part; and the straight-line edge **43c1** of one of the pair of second lead-out parts **43** and the straight-line edge **43c1** of the other one of the pair of second lead-out parts **43** have a predetermined distance ($2 \times D1$) between each other and are arranged parallel to each other. Further, the second mounting parts **42** and **42** are arranged along the base surfaces **20a2** and **20b2** of the core **20** and are arranged on opposite sides with respect to each other along the X-axis.

[0088] By configuring as such, the insulation distance between the pair of straight-line edges **43c1** can be secured easily, and even in the case that the area of the base surface of the core **20** is relatively small, the second lead-out parts **43** positioned at the both ends of the second conductor **43** can be easily insulated securely. Also, the second mounting parts **42** positioned at the both ends of the second conductor **40** can be easily insulated securely. Further, the first mounting part **32** and the second mounting part **42** can be easily insulated securely.

[0089] In the present embodiment, the core **20** includes the first core **20a** and the second core **20b** facing the first core **20a**. The second mounting parts **42** and **42** are positioned along the base surfaces **20a2** and **20b2** of the core **20** and arranged outside of the first mounting parts **32** and **32** along the X-axis parallel to the direction which the first core **20a** and the second core **20b** are facing.

[0090] By configuring as such, even in the case that the area of the base surfaces **20a2** and **20b2** of the core **20** are relatively small, the second lead-out parts **43** positioned at the both ends of the second conductor **40** can be easily insulated securely. Also, the second mounting parts **42** and **42** positioned at both ends of the second conductor **40** can be easily insulated securely. Further, the first mounting part **32** and the second mounting part **42** can be easily insulated securely.

[0091] In the present embodiment, a conductor width of the second mounting part **42** along the Y-axis is wider than the conductor width **W0** of the second lead-out part **43**. By configuring as such, bonding areas between the second mounting parts **42** and the land electrodes **54a** and **54b** of the base board **50** shown in FIG. 5 can be increased; hence, the bonding strength improves. Also, even in the case that the position shifting occurs between the second mounting parts **42** and the land electrodes **54a** and **54b** of the base board **50**, bonding strength between the land electrodes **54a** and **54b** and the second mounting parts **42** can be secured easily.

[0092] In the present embodiment, the width **W1** of the second main body **41** along the X-axis is narrower than the width of the first main body **31**. When such relationship is satisfied, the second curved part **43b** and the intermediate part **43c** can be easily formed to the second lead-out part **43**, and even in the case that the area of the base surface of the core is relatively small, the second lead-out parts **43** positioned at the both ends of the second conductor **40** can be easily insulated securely. Also, the second mounting parts **42** positioned at the both ends of the second conductor **40** can be easily insulated securely. Further, insulation between the first mounting part **32** and the second mounting part **42** can be secured easily.

[0093] Also, in the present embodiment, as shown in FIG. 5, a plurality of coil devices **10** can be

easily aligned in one line (or it may be in a plurality of lines) along the X-axis, and the mounting parts **42** of the coil devices **10** adjacent to each other can be aligned along the X-axis in a close proximity. Therefore, due to the common second land electrodes **54b** formed on the mounting surface of the circuit board **50**, it is possible to easily form a circuit having the second conductors **40** connected in series in which the second conductors **40** shown in FIG. 2A and FIG. 2B are arranged inside the plurality of coil devices **10** along the X-axis connected as shown in FIG. 5. Also, a circuit with simple configuration can be achieved.

[0094] Also, unlike as shown in FIG. 5, one coil device **10** can connect to the land electrodes **53**, **54a**, and **54b** of the base board **50** using a solder or so. For example, in such case, when viewing from the upper side along the Z-axis of the device **10**, the bonding material which is raised can be easily observed, such as a solder fillet between the outer end edge of each of the mounting parts **32** and **42** and the land electrodes **53**, **54a**, and **54b** can be observed easily. By observing the bonding material such as a solder fillet which is raised, it becomes easy to determine the condition of bonding.

[0095] Note that, as shown in FIG. 5, even in the case of connecting the plurality of coil devices **10** to the base board **50** along the X-axis, the solder fillet of the mounting part **42** positioned at the both ends along the X-axis and the solder fillet of the mounting part **32** of each coil device can be easily observed from the upper side along the Z-axis of the coil device **10**.

Second Embodiment

[0096] As shown in FIG. 3B to FIG. 3D, the coil device according to the second embodiment has basically the same configuration as the first embodiment, except that the coil device according to the second embodiment has configurations that shapes of the second lead-out part **43** and/or the second mounting part **42** of the second conductor **40** are different from those of the first embodiment. The configurations of the second embodiment which are the same as the first embodiment exhibit the same effects. In the FIG. 3B to FIG. 3D, the parts which are different from the first embodiment are only shown, and the explanation of the same configurations are omitted.

[0097] In the embodiment shown in FIG. 3B, each mounting part **42** is configured as an extension part of the intermediate part **43c**, and a width along the Y-axis of the mounting part **42** of the embodiment shown in FIG. 3B is narrower than the width along the Y-axis of the mounting part **42** shown in FIG. 3A. In the embodiment shown in FIG. 3C, a width along the Y-axis of the intermediate part **43c** is configured to gradually become wider towards each mounting part **42**. In the embodiment shown in FIG. 3D, a width along the Y-axis of the intermediate part **43c** is configured to gradually become wider towards each mounting part **42**, and also the width along the Y-axis of the mounting part **42** of the embodiment shown in FIG. 3D is wider than the width along the Y-axis of the mounting part **42** shown in FIG. 3A.

[0098] Note that, the present disclosure is not limited to the above-mentioned embodiments, and various modifications are possible within the scope of the present disclosure.

[0099] For example, in the above-mentioned embodiments, the base surfaces **20a2** and **20b2** of the core **20** are roughly a square shape; however, it may be any other polygon shape, or the base surface of the core may have at least four outer edges which configure a polygon shape. Each of the pair of first mounting parts **32** and **32** are respectively arranged facing the two outer edges which are facing away from each other among the four outer edges, and each of the pair of second mounting parts **42** may be arranged facing the outer edges facing away from each other which are different from those of the first mounting parts **32** and **32** are facing.

[0100] By configuring as such, as similar to the aforementioned embodiments, the second lead-out parts **43** positioned at the both ends of the second conductor **40** can be easily insulated securely. Also, the second mounting parts **42** positioned at the both ends of the second conductor **40** can be easily insulated securely. Further, insulation between the first mounting part **32** and the second mounting part **42** can be secured easily.

[0101] Also, the aforementioned embodiments are configured so that the pair of intermediate parts

43 respectively has the straight-line edge **43c1**. The part where the straight-line edge **43c1** is formed may be a curved-line edge, a zigzag-line edge, a wavy edge, etc. Note that, the pair of intermediate parts is preferably shaped so that the distance between the edges which are facing to each other of the pair of intermediate parts **43c** maintains the insulation distance ($2 \times D1$).

REFERENCE SIGNS LISTS

[0102] **10** . . . Coil device [0103] **20** . . . Core [0104] **20a** . . . First core [0105] **20a1** . . . Upper end surface [0106] **20a2** . . . Base surface [0107] **20a3** . . . Outer side surface [0108] **20a4** . . . Outer side surface [0109] **20b** . . . Second core [0110] **20b1** . . . Upper end surface [0111] **20b2** . . . Base surface [0112] **20b3** . . . Outer side surface [0113] **20b4** . . . Outer side surface [0114] **21a** . . . First base part [0115] **21b** . . . Second base part [0116] **22a** . . . First outer leg part [0117] **22b** . . . Second outer leg part [0118] **23a** . . . First middle leg part [0119] **23a1** . . . Upper end surface [0120] **23b** . . . Second middle leg part [0121] **24a** . . . First step-like depressed part [0122] **24b** . . . Second step-like depressed part [0123] **25a** . . . First raised part [0124] **25a1** . . . Upper end face [0125] **25b** . . . Second raised part [0126] **25b1** . . . Upper end surface [0127] **26a** . . . First groove [0128] **26b** . . . Second groove [0129] **30** . . . First conductor [0130] **31** . . . First main body [0131] **31a** . . . Leg part [0132] **31b** . . . Connecting part [0133] **32** . . . First mounting part [0134] **33** . . . Lead-out part [0135] **40** . . . Second conductor [0136] **41** . . . Second main body [0137] **41a** . . . Leg part [0138] **41b** . . . Connecting part [0139] **42** . . . Second mounting part [0140] **43** . . . Lead-out part [0141] **43a** . . . First curved part [0142] **43a1** . . . Inward bending beginning line [0143] **43b** . . . Second curved part [0144] **43c** . . . Intermediate part [0145] **43c1** . . . Straight-line edge [0146] **50** . . . Base board [0147] **53** . . . First land electrode [0148] **54a** . . . Second land electrode [0149] **54b** . . . Common second land electrode

Claims

1. A coil device comprising: a first conductor having a first main body, a second conductor having a second main body arranged inside the first main body and at least partially extending along the first main body, and a core in which the first main body and the second main body are arranged inside of the core; wherein the second main body is arranged inside of the first main body in the core and extends along the first main body, the second conductor comprises a second mounting part continuous with the second main body, a first mounting part of the first conductor and the second mounting part of the second conductor are arranged along a base surface of the core, and a second lead-out part provided between the second main body and the second mounting part comprises a curved part which changes a lead-out direction along the base surface so that the second mounting part is pulled out to a position different from the first mounting part.
2. The coil device according to claim 1, wherein the second lead-out part further comprises an intermediate part connecting the curved part and the second mounting part in a continuous manner, and the intermediate part comprises a straight-line edge which is continuous with an outline of the curved part.
3. The coil device according to claim 2, wherein the second lead-out part comprises a pair of second lead-out parts formed at both ends of the second conductor, each of the pair of the second lead-out parts respectively comprises the straight-line edge, the straight-line edge of one of the pair of second lead-out parts and the straight-line edge of the other one of the pair of second lead-out parts have a predetermined distance between each other and are arranged parallel to each other, and the second mounting part comprises a pair of second mounting parts and each of the pair of second mounting parts is arranged at a position which is opposite with respect to each other along the base surface of the core.
4. The coil device according to claim 1, wherein the core comprises a first core and a second core facing the first core; the second mounting part is positioned outside of the first mounting part along a first axis parallel to a direction which the first core and the second core are facing each other

along a base surface of the core.

5. The coil device according to claim 1, wherein a conductor width of the second mounting part is wider than a conductor width of the second lead-out part.

6. The coil device according to claim 1, wherein the first mounting part comprises a pair of first mounting parts formed at both ends of the first conductor, the second mounting part comprises a pair of second mounting parts formed at both ends of the second conductor, a base surface of the core comprises at least four outer edges configuring a polygon shape, each of the pair of first mounting parts is arranged in a direction facing two of the at least four outer edges facing away from each other, and each of the pair of second mounting parts is arranged in a direction facing two of the at least four outer edges facing away from each other which are different edges that the first mounting part are facing.

7. The coil device according to claim 1, wherein the first conductor and the second conductor respectively has a flat wire shape within one turn.

8. The coil device according to claim 1, wherein a width of the second main body is narrower than a width of the first main body.

9. The coil device according to claim 8, wherein an inner surface of the core comprises a step part corresponding to each width of the second main body and the first main body.

10. The coil device according to claim 1, wherein an insulation layer is formed on a surface of the first main body.
